

数学练习 第 04 份 答案与解析

共 83 题 与《练习04》题号一一对应

1. A 中 C 可以不可逆, 只能保证半正定; B 中负惯性指数为 0 仍可能有零特征值; C 中 $P^{-1}AP = E$ 意味 $A = E$, 过强.

$D: A$ 正定 $\Rightarrow A^*$ 的特征值 $\frac{|A|}{\lambda_i} > 0$, A^* 正定, 与 E 合同; 反之若 A^* 正定, 设 A 的特征值全为负, 则 n 为偶数时 $|A| > 0$, $\frac{|A|}{\lambda_i} < 0$ 矛盾, 故 A 正定. 选 D .

2. 两边取对数: $\frac{1}{2} \ln(x^2 + y^2) = \arctan \frac{y}{x}$, 求导得

$$\frac{x + yy'}{x^2 + y^2} = \frac{xy' - y}{x^2 + y^2} \Rightarrow y' = \frac{x + y}{x - y}.$$

再求导:

$$y'' = \frac{(1 + y')(x - y) - (x + y)(1 - y')}{(x - y)^2} = \frac{2(x^2 + y^2)}{(x - y)^3}.$$

3. $P(A|B) = P(A)$ 表明 A, B 相互独立, 从而 $P(AB) = P(A)P(B) > 0$, A, B 必相容, 故"互不相容"不正确. 选 A .

4. 梯度场沿任意闭曲线的环流为零 ($\text{rot grad} \equiv 0$), 故 $I = 0$.

5. L 上点为 $(1 + 3t, 2 + 4t, -1 + t)$, 转轴为过 $(2, 3)$ 平行 z 轴的直线. 由 $z = -1 + t$ 得 $t = z + 1$, 旋转时保持到轴的距离不变:

$$(x - 2)^2 + (y - 3)^2 = (3t - 1)^2 + (4t - 1)^2 = (3z + 2)^2 + (4z + 3)^2,$$

即 $(x - 2)^2 + (y - 3)^2 = 25z^2 + 36z + 13$.

6. $EX_i = 0$, $DX_i = EX_i^2 = 2 \cdot \frac{(ia)^2}{2i^2} = a^2$, 各 X_i 独立且方差一致有界.

$$D\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \cdot na^2 = \frac{a^2}{n},$$

由切比雪夫不等式

$$P\left\{\left|\frac{1}{n} \sum X_i\right| \geq \varepsilon\right\} \leq \frac{a^2}{n\varepsilon^2} \rightarrow 0 \quad (n \rightarrow \infty).$$

7. 令 $u = \tan y$, 方程化为 $u' + \frac{x}{1+x^2}u = x$, 积分因子 $\sqrt{1+x^2}$:

$$(\sqrt{1+x^2}u)' = x\sqrt{1+x^2} \Rightarrow u = \frac{1+x^2}{3} + \frac{C}{\sqrt{1+x^2}}.$$

由 $y(0) = 0$ 得 $u(0) = 0$, $C = -\frac{1}{3}$, 故

$$\tan y = \frac{1+x^2}{3} - \frac{1}{3\sqrt{1+x^2}}, \quad y = \arctan \left[\frac{1+x^2}{3} - \frac{1}{3\sqrt{1+x^2}} \right].$$

8. 由 $P^{-1}A^{-1}P = B^{-1}$ 取逆得 $P^{-1}AP = B$, 故 ② 成立; 由 $A \sim B$ 立得 ③、④ 成立; 又 A 可逆时 $AB = A(BA)A^{-1}$, 故 ① 成立. 四个都对, 选 D .

9. 逐项求导不改变收敛半径, $\sum na_n x^n$ 的收敛半径仍为 3, 故

$$\sum_{n=0}^{\infty} na_n(x+1)^{n+1} = (x+1) \sum_{n=0}^{\infty} na_n(x+1)^n$$

的收敛区间为 $|x+1| < 3$, 即 $(-4, 2)$.

10. 利用分块初等变换:

$$\begin{pmatrix} O & A \\ B & E \end{pmatrix} \rightarrow \begin{pmatrix} -AB & O \\ O & E \end{pmatrix}, \quad \begin{pmatrix} A & B \\ O & E \end{pmatrix} \rightarrow \begin{pmatrix} A & O \\ O & E \end{pmatrix}, \quad \begin{pmatrix} A & AB \\ E & B \end{pmatrix} \rightarrow \begin{pmatrix} A & O \\ E & O \end{pmatrix},$$

故 $r_1 = n + r(AB)$, $r_2 = n + r(A)$, $r_3 = n$. 又 $r(AB) \leq r(A)$,

所以 $r_2 \geq r_1 \geq r_3$. 选 A.

11. (X, Y) 的密度为 $\frac{1}{2}$. 当 $0 \leq z \leq 2$ 时

$$F_Z(z) = \frac{1}{2} \left[z \cdot 1 + \int_z^2 \frac{z}{x} dx \right] = \frac{1}{2} \left[z + z \ln \frac{2}{z} \right],$$

$z < 0$ 时为 0, $z > 2$ 时为 1. 求导得

$$f_Z(z) = \begin{cases} \frac{1}{2} \ln \frac{2}{z}, & 0 < z < 2, \\ 0, & \text{其他.} \end{cases}$$

12. D 的面积为 2π , 形心为 $(1, 1)$, 故 $\iint_D (x+y) d\sigma = 4\pi$. 由格林公式 (顺时针取负号)

$$I_1 = - \iint_D \frac{x+y}{4} d\sigma = -\pi, \quad I_2 = - \iint_D \frac{x-y}{4} d\sigma = 0, \quad I_3 = + \iint_D \frac{x+y}{4} d\sigma = \pi.$$

故 $I_3 > I_2 > I_1$, 选 B.

13. $f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{\arcsin |x|}{x} = \pm 1$ 不存在, $f'_y(0, 0) = \lim_{y \rightarrow 0} \frac{\arcsin y^2}{y} = 0$.

选 B.

14. 由 $|A| = 0$ 及 $A_{11} \neq 0$ 知 $r(A) = n-1$, 故 $r(A^*) = 1$, $A^*x = 0$ 的解空间维数为 $n-1$.

又 $A^*A = |A|E = O$, 说明 A 的各列都是 $A^*x = 0$ 的解, 而 $r(A) = n-1$ 恰等于解空间维数, 故解空间即 A 的列空间. 因 $A_{11} \neq 0$, A 的后 $n-1$ 个列 $\alpha_2, \dots, \alpha_n$ 线性无关, 可作基础解系:

$$x = k_2\alpha_2 + k_3\alpha_3 + \dots + k_n\alpha_n.$$

15. $(1 + \frac{1}{k})^{k^2} = e^{k^2 \ln(1 + \frac{1}{k})} = e^{k - \frac{1}{2} + o(1)}$, 故通项 $\sim e^{-\frac{1}{2}} (\frac{e}{3})^k$.

因 $\frac{e}{3} < 1$, 级数 $\sum_{k=1}^{\infty} \frac{1}{3^k} (1 + \frac{1}{k})^{k^2}$ 收敛, 设和为 S , 则

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n (\dots) = \lim_{n \rightarrow \infty} \frac{S_n}{n} = 0.$$

16. 由同分布连续性, X_n 是 n 个变量中最小者的概率为 $\frac{1}{n}$, 故

$$P\{X_n > \min(X_1, \dots, X_{n-1})\} = 1 - \frac{1}{n} = \frac{n-1}{n}.$$

17. (I) 切线在 y 轴截距为 $y - xy'$, 由题意 $\sqrt{x^2 + y^2} = y - xy'$. 令 $u = \frac{y}{x}$:

$$xu' = -\sqrt{1+u^2} \implies u + \sqrt{1+u^2} = \frac{C}{x} \implies y + \sqrt{x^2 + y^2} = C.$$

由过点 $(\frac{1}{2}, 0)$ 得 $C = \frac{1}{2}$, 化简得 $L: y = \frac{1}{4} - x^2$.

(II) 设切点为 $(a, \frac{1}{4} - a^2)$ ($0 < a < \frac{1}{2}$), 切线 $y = \frac{1}{4} + a^2 - 2ax$, 与两轴围成三角形面积

$$\Delta(a) = \frac{(a^2 + \frac{1}{4})^2}{4a}, \quad \Delta'(a) = \frac{(a^2 + \frac{1}{4})(3a^2 - \frac{1}{4})}{4a^2},$$

故 $a = \frac{\sqrt{3}}{6}$ 时最小, $\Delta = \frac{\sqrt{3}}{18}$. 又曲线与两轴围成的面积为 $\int_0^{1/2} (\frac{1}{4} - x^2) dx = \frac{1}{12}$ (与 a 无关), 所求最小面积为 $\frac{\sqrt{3}}{18} - \frac{1}{12}$, 对应切线

$$y = \frac{1}{3} - \frac{\sqrt{3}}{3}x.$$

18. $x \rightarrow 0^+$: $\frac{2+e^{1/x}}{1+e^{2/x}} \rightarrow 0$, $\frac{\sin x}{|x|} \rightarrow 1$, 故 $f(0^+) = 1$;

$x \rightarrow 0^-$: $\frac{2+0}{1+0} = 2$, $\frac{\sin x}{|x|} \rightarrow -1$, 故 $f(0^-) = 1$.

左右极限相等而 f 在 0 无定义, 为可去间断点. 选 A.

19. (I) A 的特征值为 $2, -1, -1$, $|A| = 2$; 由 $A^* \alpha = \alpha$ 及 $A^* = 2A^{-1}$ 得 $A\alpha = 2\alpha$, 即 $\alpha = (1, 1, -1)^T$ 是 $\lambda = 2$ 的特征向量. 取

$$Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}, \quad Q^T A Q = \text{diag}(2, -1, -1),$$

$$A = 3 \cdot \frac{\alpha \alpha^T}{3} - E = \alpha \alpha^T - E = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}.$$

(II) $A + 2E$ 的特征值为 $4, 1, 1$, 正定平方根为

$$B = \frac{\alpha \alpha^T}{3} + E = \frac{1}{3} \begin{pmatrix} 4 & 1 & -1 \\ 1 & 4 & -1 \\ -1 & -1 & 4 \end{pmatrix}.$$

20. f 为奇函数时 $\int_0^x f(t) dt$ 为偶函数, 任一原函数 $F(x) = \int_0^x f + C$ 仍为偶函数, A 正确. B 中加常数后未必是奇函数; C 取 $f \equiv 1$; D 取 $f(x) = x$ ($F = \frac{x^2}{2}$ 非单调) 均为反例. 选 A.

21. 三直线恰交于一点 $\iff$ 线性方程组 $(a_i \ b_i) \begin{pmatrix} x \\ y \end{pmatrix} = -c_i$ 有唯一解 $\iff r(\alpha_1, \alpha_2) = r(\alpha_1, \alpha_2, \alpha_3) = 2$. 选 D.

22. 极坐标: $0 \leq \theta \leq \frac{\pi}{4}$, $2 \cos \theta \leq r \leq \frac{2}{\cos \theta}$,

$$I = \int_0^{\frac{\pi}{4}} \int_{2 \cos \theta}^{\frac{2}{\cos \theta}} dr d\theta = 2 \int_0^{\frac{\pi}{4}} (\sec \theta - \cos \theta) d\theta = 2 \ln(1 + \sqrt{2}) - \sqrt{2}.$$

23. S 是球心 $(1, 0, 0)$ 、半径 1 的球面, 其上 $\rho = x^2 + y^2 + z^2 = 2x$,

$$m = \iint_S 2x dS = 2\bar{x} \cdot 4\pi = 2 \cdot 1 \cdot 4\pi = 8\pi.$$

24. $P\{X = k\} = \left(\frac{1}{4}\right)^{k-1} \cdot \frac{3}{4},$

$$\sum_{n=1}^{\infty} P\{X = 2n\} = \frac{3}{4} \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^{2n-1} = \frac{3}{4} \cdot \frac{\frac{1}{4}}{1 - \frac{1}{16}} = \frac{1}{5}.$$

25. $F = x^2 + y^2 - 1 - z, \nabla F|_P = (4, 2, -1).$

切平面: $4(x-2) + 2(y-1) - (z-4) = 0$, 即 $4x + 2y - z - 6 = 0$;

法线: $\frac{x-2}{4} = \frac{y-1}{2} = \frac{z-4}{-1}.$

26. 以 2 为周期 ($l=1$), 在 $[1, 3]$ 上计算:

$$\frac{a_0}{2} = \frac{1}{2} \int_1^3 f(x) dx = \frac{3}{4}, \quad a_n = \int_1^3 f \cos n\pi x dx = \frac{1 - (-1)^n}{n^2 \pi^2}, \quad b_n = \int_1^3 f \sin n\pi x dx = \frac{(-1)^n}{n\pi}.$$

故

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \left[\frac{1 - (-1)^n}{n^2 \pi^2} \cos n\pi x + \frac{(-1)^n}{n\pi} \sin n\pi x \right],$$

在 f 的连续点处成立; 在 $x=1, 3$ (及其同余点) 处级数收敛到 $\frac{1+0}{2} = \frac{1}{2}.$

27. $|A| = |E + \alpha\beta^T| = 1 + \beta^T\alpha = 1 + 2 = 3 \neq 0$, 故 A 可逆.

设 $A^{-1} = E + k\alpha\beta^T$, 由

$$(E + \alpha\beta^T)(E + k\alpha\beta^T) = E + (1 + k + 2k)\alpha\beta^T = E$$

得 $k = -\frac{1}{3}$, 即 $A^{-1} = E - \frac{1}{3}\alpha\beta^T.$

28. (原题应附有条件 $|f''(x)| \leq 1$, 否则结论不成立: 取 $f(x) = c + M \sin 4\pi x$ 满足全部已知条件而 $|f'(0)| + |f'(1)| = 8\pi|M|$ 可任意大. 下面在 $|f''| \leq 1$ 下证明.)

由积分中值定理, 存在 $\xi \in [\frac{1}{2}, 1]$ 使 $2 \int_{\frac{1}{2}}^1 f dx = f(\xi)$, 故 $f(0) = f(\xi) = f(1).$

对 $[0, \xi]$ 、 $[\xi, 1]$ 分别用罗尔定理, 得 $c \in (0, \xi)$ 、 $d \in (\xi, 1)$ 使 $f'(c) = f'(d) = 0$. 于是

$$|f'(0)| = \left| \int_0^c f'' dx \right| \leq c, \quad |f'(1)| = \left| \int_d^1 f'' dx \right| \leq 1 - d,$$

相加得 $|f'(0)| + |f'(1)| \leq c + 1 - d < 1$ ($\xi = 1$ 时直接对 $[0, 1]$ 用罗尔定理同理).

29. 齐次解 $(C_1 + C_2 x)e^{-x}$; 设特解 $y^* = (ax + b)e^x$, 代入得 $4ax + 4a + 4b = x$, $a = \frac{1}{4}$, $b = -\frac{1}{4}.$

$$y = (C_1 + C_2 x)e^{-x} + \frac{x-1}{4}e^x; \quad y(0) = 0 \Rightarrow C_1 = \frac{1}{4}, \quad y'(0) = 0 \Rightarrow C_2 = \frac{1}{4}.$$

故 $y = \frac{1+x}{4}e^{-x} + \frac{x-1}{4}e^x.$

30. (I) $A^T = A$ 为实对称矩阵, 必可正交相似对角化. 具体地

$$A(\alpha + \beta) = \alpha + \beta, \quad A(\alpha - \beta) = -(\alpha - \beta), \quad A\gamma = 0 \quad (\gamma \perp \alpha, \beta),$$

特征值为 $1, -1, 0$, 三者互异, 故 A 相似于对角矩阵.

(II) 由上式知 $P = (\gamma, 2(\alpha + \beta), \beta - \alpha)$ 的三列分别是特征值 $0, 1, -1$ 的特征向量, 故

$$P^{-1}AP = \text{diag}(0, 1, -1).$$

31. 由 $\lim_{x \rightarrow 0} \frac{f(x)}{1 - \cos x} = 2$ 知在 0 的去心邻域内 $f(x) > 0 = f(0)$, 故 f 在 $x = 0$ 取极小值;

又 $\frac{f(x)}{x} = \frac{f(x)}{1 - \cos x} \cdot \frac{1 - \cos x}{x} \rightarrow 2 \cdot 0 = 0$, 故 $f'(0) = 0$ 存在. 选 D.

$$32. \quad V = \iint_{r \leq h} \left(h - \frac{r^2}{h}\right) d\sigma = \frac{\pi h^3}{2}, \quad S = \iint_{r \leq h} \sqrt{1 + \frac{4r^2}{h^2}} d\sigma = \frac{\pi h^2}{6} (5\sqrt{5} - 1).$$

由 $\frac{dV}{dt} = -aS$ 得

$$\frac{3\pi h^2}{2} \frac{dh}{dt} = -a \frac{\pi h^2}{6} (5\sqrt{5} - 1) \implies \frac{dh}{dt} = -\frac{a(5\sqrt{5} - 1)}{9},$$

故 $h(t) = h_0 - \frac{a(5\sqrt{5} - 1)}{9}t$, 全部融化所需时间

$$T = \frac{9h_0}{a(5\sqrt{5} - 1)}.$$

33. (I) $\iint_{\mathbf{R}^2} (1 + xy) e^{-\frac{x^2+y^2}{2}} dx dy = 2\pi + 0 = 2\pi$, 故 $k = 2\pi$.

边缘密度 $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, 同理 $f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$,

而 $f(x, y) \neq f_X(x)f_Y(y)$, 故 X 与 Y 不独立 (但都服从 $N(0, 1)$).

(II) 由卷积公式

$$f_U(u) = \int_{-\pi}^{\pi} \frac{1}{2\pi} \varphi(u - z) dz = \frac{\Phi(u + \pi) - \Phi(u - \pi)}{2\pi}.$$

$$34. \quad \cos x - e^{\frac{x^2}{2}} = -x^2 + o(x^2), \quad \sin x^2 \sim x^2, \quad \frac{x^2}{2} + 1 - \sqrt{1 + x^2} = \frac{x^4}{8} + o(x^4),$$

故极限 $= \frac{-x^4}{x^4/8} = -8$.

35. (II) 的解为 $k_1\alpha_1 + k_2\alpha_2 = (-k_1, 2k_1 - k_2, 2k_1 - k_2, k_1)^T$, 代入 (I) :

$$x_1 + x_2 = k_1 - k_2 = 0, \quad x_2 - x_4 = k_1 - k_2 = 0,$$

故 $k_1 = k_2 = k$, 非零公共解为

$$k(\alpha_1 + \alpha_2) = k(-1, 1, 1, 1)^T, \quad k \neq 0.$$

$$36. \quad f(x) = \int_0^x \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{n!} dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)}, \quad -\infty < x < +\infty.$$

37. 对 $g(x+y)$ 型函数, $\iint_D g(x+y)d\sigma = \int_0^t u g(u)du$; 又由 $u = x+y, v = x-y$ 知 $\iint_D (x-y)^3 d\sigma = 0$

故条件化为

$$\int_0^t u f''(u)du = \int_0^t u f(u)du \quad (\forall t),$$

求导得 $tf''(t) = tf(t)$, 即 $f'' = f$. 于是 $f(x) = C_1 e^x + C_2 e^{-x}$,

由 $f(0) = 1, f'(0) = 1$ 得 $C_1 = 1, C_2 = 0$, 故 $f(x) = e^x$.

38. 在 L 上分母恒为 1, 故 $I = \oint_L (-y dx + x dy) = 2S$, S 为 L 所围面积.

令 $u = \ln x, v = \ln y$, 则 L 变为 $|u| + |v| = 1, dx dy = e^{u+v} du dv$,

$$S = \iint_{|u|+|v|\leq 1} e^{u+v} du dv = \frac{1}{2} \int_{-1}^1 \int_{-1}^1 e^s dd ds = e - \frac{1}{e},$$

(其中 $s = u+v, d = u-v$) . 故 $I = 2\left(e - \frac{1}{e}\right)$.

39. F 是两个正态分布函数的混合,

$$EX = 0.3 \times 4 + 0.7 \times (-1) = 0.5.$$

40. 两曲面交于 $r = 1, z = 0$. 用柱面坐标, $r-1 \leq z \leq \sqrt{1-r^2} (0 \leq r \leq 1)$:

$$I = 2\pi \int_0^1 \frac{r}{3} \left[(1-r^2)^{\frac{3}{2}} - (r-1)^3 \right] dr = \frac{2\pi}{3} \left(\frac{1}{5} + \frac{1}{20} \right) = \frac{\pi}{6}.$$

41. 两曲面法向量 $\mathbf{n}_1 = (6, 2, 0), \mathbf{n}_2 = (6, 0, 2)$, 切向量 $\mathbf{n}_1 \times \mathbf{n}_2 = (4, -12, -12) \parallel (1, -3, -3)$.

切线: $\frac{x-3}{1} = \frac{y-1}{-3} = \frac{z-1}{-3}$; 法平面: $(x-3) - 3(y-1) - 3(z-1) = 0$, 即 $x - 3y - 3z + 3 = 0$

42. 令 $u = x^2 - t^2$, 则 $F(x) = \frac{1}{2} \int_0^{x^2} f(u)du$, 于是

$$\lim_{x \rightarrow 0} \frac{F(x)}{x^4} = \frac{1}{2} \lim_{s \rightarrow 0^+} \frac{\int_0^s f(u)du}{s^2} = \frac{1}{2} \cdot \lim_{s \rightarrow 0^+} \frac{f(s)}{2s} = \frac{f'(0)}{4} = \frac{1}{4}.$$

43. 由 $AB = B^{-1}A^{-1} = (AB)^{-1}$ 得 $(AB)^2 = E$, 同理 $(BA)^2 = E$, 即 $(E+BA)(E-BA) = O$.

故 $r(E+BA) + r(E-BA) \leq n$; 又

$$r(E+BA) + r(E-BA) \geq r[(E+BA) + (E-BA)] = r(2E) = n.$$

故和为 n .

44. $y = \frac{4}{9}x^2, \frac{dy}{dt} = \frac{8}{9}x \frac{dx}{dt} \Big|_{x=3} = \frac{8}{3} \cdot 30 = 80 \text{ (cm/s)}$.

$$S = \sqrt{x^2 + y^2} = 5,$$

$$\frac{dS}{dt} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{S} = \frac{3 \cdot 30 + 4 \cdot 80}{5} = 82 \text{ cm/s}.$$

45. 隐函数 $y' = -\frac{F'_x}{F'_y}$, 故 $y'(x_0) = 0$; 再求导并代入 $y'(x_0) = 0$ 得

$$y''(x_0) = -\frac{F''_{xx}(x_0, y_0)}{F'_y(x_0, y_0)} > 0,$$

故 $y(x)$ 在 x_0 处取得极小值. 选 A.

46. 球面在柱体内的部分 (上下两块) :

$$S_1 = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{R \cos \theta} \frac{R}{\sqrt{R^2 - r^2}} r dr d\theta = 2R^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - |\sin \theta|) d\theta = 2R^2(\pi - 2).$$

柱面在球内部分: 取 $x = \frac{R}{2}(1 + \cos \varphi)$, $y = \frac{R}{2} \sin \varphi$, 则 $|z| \leq R \left| \sin \frac{\varphi}{2} \right|$, $ds = \frac{R}{2} d\varphi$,

$$S_2 = \int_0^{2\pi} 2R \left| \sin \frac{\varphi}{2} \right| \cdot \frac{R}{2} d\varphi = 4R^2.$$

故 $\frac{S_1}{S_2} = \frac{\pi - 2}{2}$.

47. $A = 3E - 2J$ (J 为全 1 矩阵), 特征值为 $3 - 2 \cdot 3 = -3$ 与 3 (二重),
故 $f = 3y_1^2 + 3y_2^2 - 3y_3^2$, $f = 1$ 表示单叶双曲面. 选 C.

48. $\bar{X} - \bar{Y} \sim N(0, \frac{2\sigma^2}{n})$,

$$P\{|\bar{X} - \bar{Y}| > \sigma\} = P\left\{|Z| > \sqrt{\frac{n}{2}}\right\}$$

与 σ 无关. 选 C.

49. 设平面束 $3x - 2y + 2 + \lambda(x - 2y - z + 6) = 0$, 即 $(3 + \lambda)x - (2 + 2\lambda)y - \lambda z + (2 + 6\lambda) = 0$.
由点 $P(1, 2, 1)$ 到平面距离为 1:

$$\frac{|1 + 2\lambda|}{\sqrt{6\lambda^2 + 14\lambda + 13}} = 1 \implies \lambda^2 + 5\lambda + 6 = 0 \implies \lambda = -2 \text{ 或 } -3.$$

故 $\pi: x + 2y + 2z - 10 = 0$ 或 $4y + 3z - 16 = 0$.

50. $y > 0$ 时 $F_Y(y) = P\{X \geq e^{-y/2}\} = 1 - e^{-y/2}$, 故

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-y/2}, & y > 0, \\ 0, & y \leq 0, \end{cases}$$

即 Y 服从参数为 $\frac{1}{2}$ 的指数分布.

- 51.

$$f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n+2} - 1}{x^{2n} + 1} = \begin{cases} x^2, & |x| > 1, \\ -1, & 0 < |x| < 1, \\ 0, & x = \pm 1. \end{cases}$$

($x = 0$ 时原式无意义.) 故 f 在 $x = \pm 1$ 处左右极限为 -1 与 1 , 是跳跃间断点;
 $x = 0$ 处极限为 -1 , 是可去间断点; 其余各点连续.

52. 对 $(\alpha_1, \alpha_2, \alpha_3, \alpha_4 | \alpha)$ 作行变换得

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 0 \\ 0 & a+1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 2a-1 & 1 \\ 0 & 0 & 0 & 0 & b-1 \end{pmatrix}.$$

(I) 当 $a \neq \frac{1}{2}$ 时 $r(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = 3$, 极大无关组可取 $\alpha_1, \alpha_2, \alpha_4$ (当 $a = -1$ 时取 $\alpha_1, \alpha_3, \alpha_4$); 当 $a = \frac{1}{2}$ 时秩为 2, 极大无关组为 α_1, α_2 .

(II) α 不能被线性表示 $\iff$ 增广矩阵秩更大:

当 $a = \frac{1}{2}$ 时 (第三行成为 $0 = 1$) 对任意 b 都不能表示; 当 $a \neq \frac{1}{2}$ 时须 $b \neq 1$.

53. (I) $\frac{1}{x^2-3x+2} = \frac{1}{1-x} - \frac{1}{2} \cdot \frac{1}{1-\frac{x}{2}} = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) x^n, |x| < 1.$

(II) $\ln(1-x-2x^2) = \ln(1-2x) + \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}-2^n}{n} x^n$, 收敛域 $[-\frac{1}{2}, \frac{1}{2})$.

(III) $\ln(x + \sqrt{1+x^2}) = \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}$, 收敛域 $[-1, 1]$.

(IV) $x \arctan x - \frac{1}{2} \ln(1+x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)(2n+2)} x^{2n+2}$, 收敛域 $[-1, 1]$.

54. 分离变量: $\frac{dx}{2x-1} = -\frac{y dy}{1+y^2}$, 积分得 $\frac{1}{2} \ln|2x-1| = -\frac{1}{2} \ln(1+y^2) + C$, 即

$$(2x-1)(1+y^2) = C.$$

55. (I) $A^2 = A$ 且实对称, 特征值只能为 0, 1, 由 $r(A) = r$ 知 1 为 r 重、0 为 $n-r$ 重,

$$|3E - A| = 2^r \cdot 3^{n-r}.$$

(II) 一般矩阵满足 $A^2 = A$ 时最小多项式 $\lambda^2 - \lambda$ 无重根, A 仍可对角化, 结论相同: $|3E - A| = 2^r 3^{n-r}$.

56. 以第二个方程为主元作消元:

$$\begin{pmatrix} 1 & 0 & 1 & -1 & -3 \\ 0 & -1 & 2 & -1 & 2 \\ 0 & 1 & -2 & 3 & 10 \\ 0 & 0 & 0 & 4 & 24 \end{pmatrix} \Rightarrow x_4 = 6, x_2 = 2x_3 - 8, x_1 = 3 - x_3.$$

故通解

$$x = (3, -8, 0, 6)^T + t(-1, 2, 1, 0)^T, \quad t \in \mathbf{R}.$$

57. 增加最快的方向即梯度方向

$$\nabla f(1, -2) = (1, -1), \quad \text{单位方向} \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right).$$

58. 由对称性 $\iint_D x^2 d\sigma = \iint_D y^2 d\sigma = \frac{1}{2} \iint_D r^2 d\sigma = \frac{\pi}{4}$, 故

$$I = \left(\frac{1}{4} + \frac{1}{9} \right) \cdot \frac{\pi}{4} = \frac{13\pi}{144}.$$

59. $\bar{X} \sim N\left(\mu, \frac{8}{36}\right)$, $\sigma_{\bar{X}} = \frac{\sqrt{2}}{3}$,

$$P\{|\bar{X} - \mu| < 1\} = 2\Phi\left(\frac{1}{\sqrt{2}/3}\right) - 1 = 2\Phi\left(\frac{3\sqrt{2}}{2}\right) - 1 \approx 0.966.$$

60. 补上从 $C(1,0)$ 到 $A(1,1)$ 的直线段 $\overline{CA}$, 所得闭曲线为逆时针且内部含原点, 故

$$\oint = 2\pi.$$

而在 $\overline{CA}$ 上 $x=1, dx=0: \int_{\overline{CA}} = \int_0^1 \frac{dy}{1+y^2} = \frac{\pi}{4}$.

$$\text{故 } I = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}.$$

61. 由切比雪夫不等式的形式知 $EX = \frac{b+1}{2} = 1$, 故 $b=3$, 此时 $DX = \frac{(b+1)^2}{12} = \frac{4}{3}$.
由 $1 - \frac{DX}{\varepsilon^2} = \frac{2}{3}$ 得 $\varepsilon^2 = 4$, 即 $\varepsilon = 2$.

62. (I) $y' = \sqrt{\sin \frac{x}{n}}$,

$$S_n = \int_0^{n\pi} \sqrt{1 + \sin \frac{x}{n}} dx = n \int_0^\pi \left(\sin \frac{u}{2} + \cos \frac{u}{2} \right) du = 4n.$$

(II) $\sum \frac{x^n}{S_n S_{n+1}} = \frac{1}{16} \sum \frac{x^n}{n(n+1)}$, 收敛域 $[-1, 1]$. 由 $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$:

$$S(x) = \frac{1}{16} \left[1 + \frac{1-x}{x} \ln(1-x) \right] \quad (x \neq 0, x \neq 1), \quad S(0) = 0, \quad S(1) = \frac{1}{16}.$$

63. 两边求导: $f'(x) = -\sin x - \int_0^x f(t)dt$, 再求导得 $f'' + f = -\cos x$, 且 $f(0) = 1, f'(0) = 0$.
齐次解 $C_1 \cos x + C_2 \sin x$, 共振特解 $-\frac{x}{2} \sin x$, 代入初值得 $C_1 = 1, C_2 = 0$:

$$f(x) = \cos x - \frac{x}{2} \sin x.$$

64. $x \rightarrow +\infty$ 时 $\sqrt{ax^2 - x + 3} \sim \sqrt{a}x$, 由极限存在得 $\sqrt{a} = 2, a = 4$; 再由

$$\sqrt{4x^2 - x + 3} - 2x = \frac{-x + 3}{\sqrt{4x^2 - x + 3} + 2x} \rightarrow -\frac{1}{4} = b.$$

故斜渐近线为 $y = 2x - \frac{1}{4}$.

65. $E(X + Y - 2Z) = 1 + 1 + 2 = 4$;

$$D(X + Y + Z) = 3 + 2[\text{Cov}(X, Y) + \text{Cov}(X, Z) + \text{Cov}(Y, Z)] = 3 + 2\left(0 + \frac{1}{2} - \frac{1}{2}\right) = 3.$$

66. $|A| = -2, A^* = |A|A^{-1} = \text{diag}(-2, 1, -2)$. 由 $(A^* - 2E)BA = -8E$ 得

$$BA = -8 \text{diag}(-4, -1, -4)^{-1} = \text{diag}(2, 8, 2),$$

$$B = \text{diag}(2, 8, 2)A^{-1} = \text{diag}(2, 8, 2) \text{diag}\left(1, -\frac{1}{2}, 1\right) = \text{diag}(2, -4, 2).$$

67. 由 f 为偶函数,

$$F(-x_0) = \int_{-\infty}^{-x_0} f = \int_{x_0}^{+\infty} f = a - F(x_0) = a - \left[\frac{a}{2} + \int_0^{x_0} f\right] = \frac{a}{2} - \int_0^{x_0} f(x)dx.$$

选 C.

68. (I) $-f'_x = x^2 - y$, $-f'_y = 1 - x$, 即 $f'_x = y - x^2$, $f'_y = x - 1$, 积分得

$$f(x, y) = xy - \frac{x^3}{3} - y + C,$$

由 $f(1, 1) = -\frac{1}{3}$ 得 $C = 0$, 故 $f(x, y) = xy - \frac{x^3}{3} - y$.

(II) 内部驻点 $(1, 1)$, $f = -\frac{1}{3}$; 边界 $y = 0$ 时 $f = -\frac{x^3}{3} \leq 0$; $x = 0$ 时 $f = -y \leq 0$; 边界 $x + y = 7$ 上 $f = -\frac{x^3}{3} - x^2 + 8x - 7$, $f' = -x^2 - 2x + 8 = 0 \Rightarrow x = 2$, $f(2, 5) = \frac{7}{3}$. 故最大值为 $\frac{7}{3}$ (在 $(2, 5)$ 处取得).

69. (I) $P\{Z \leq -\frac{1}{2} | X = 0\} = P\{Y \geq \frac{1}{2}\} = e^{-\frac{1}{2}}$.

(II) $F_Z(z) = 0.4P\{Y \geq -z\} + 0.6P\{Y \geq 1 - z\}$, 即

$$F_Z(z) = \begin{cases} (0.4 + 0.6e^{-1})e^z, & z < 0, \\ 0.4 + 0.6e^{z-1}, & 0 \leq z < 1, \\ 1, & z \geq 1. \end{cases}$$

70. 上侧时 $dydz = (-z'_x)dxdy = 2x dxdy$, $dzdx = (-z'_y)dxdy = 2y dxdy$, 故

$$I = \iint_D [2x^2z + 2xy^2 + yz]dxdy, \quad D: x^2 + y^2 \leq 1, x, y \geq 0, z = 2 - r^2.$$

用极坐标分别算得 $\frac{\pi}{6}$ 、 $\frac{2}{15}$ 、 $\frac{7}{15}$, 故

$$I = \frac{\pi}{6} + \frac{2}{15} + \frac{7}{15} = \frac{\pi}{6} + \frac{3}{5}.$$

71. 由 $\alpha_1 + 2\alpha_2 - \alpha_3 = 0$ 知 $A\xi = 0$, $\xi = (1, 2, -1)^T$, 即 0 是特征值; 又 1 是二重特征值, 故特征值为 1, 1, 0.

(I) $\lambda = 1$ 的特征子空间为 $\xi^\perp$, 取其中正交基 $(2, -1, 0)^T, (1, 2, 5)^T$, 单位化后与 $\frac{\xi}{\sqrt{6}}$ 合成正交矩阵

$$Q = \left(\frac{(1, 2, -1)^T}{\sqrt{6}}, \frac{(2, -1, 0)^T}{\sqrt{5}}, \frac{(1, 2, 5)^T}{\sqrt{30}} \right),$$

则 $X = QY$ 把 f 化为标准形 $y_2^2 + y_3^2$.

(II) A^* 的特征值为 $1 \cdot 1 = 1$ (对应 ξ) 与 0, 0 (对应 $\xi^\perp$), 故 $A^*X = 0$ 的解空间为 $\xi^\perp$:

$$X = k_1(-2, 1, 0)^T + k_2(1, 0, 1)^T.$$

72. 令 $t = x - 2$:

$$f = (t + 2)e^{t+2} = e^2 \sum_{n=0}^{\infty} \frac{n+2}{n!} t^n = \sum_{n=0}^{\infty} \frac{e^2(n+2)}{n!} (x-2)^n \quad (-\infty < x < +\infty),$$

$$\text{故 } f^{(n)}(2) = n! \cdot \frac{e^2(n+2)}{n!} = (n+2)e^2.$$

73. 令 $u = \frac{y}{x} : xu' = -\frac{1+u^2}{1+u}$, 分离变量得

$$\frac{1}{2} \ln(1+u^2) + \arctan u = -\ln|x| + C \implies \frac{1}{2} \ln(x^2+y^2) + \arctan \frac{y}{x} = C.$$

由 $y(1) = 0$ 得 $C = 0$, 故特解为

$$\ln(x^2+y^2) + 2 \arctan \frac{y}{x} = 0, \quad \text{即} \quad \sqrt{x^2+y^2} = e^{-\arctan \frac{y}{x}}.$$

74. $P(AB) = 0.12, P(AC) = 0, P(BC) = P(B|C)P(C) = 0.08$.

(I) $P(A \cup B) = 0.4 + 0.3 - 0.12 = 0.58$.

(II) $P(C|A \cup B) = \frac{P(AC) + P(BC) - P(ABC)}{0.58} = \frac{0.08}{0.58} = \frac{4}{29}$.

(III) 由 $A \subset \overline{C}$ 得 $P(AB\overline{C}) = P(AB) = 0.12$, 故 $P(AB|\overline{C}) = \frac{0.12}{0.6} = 0.2$.

75. ① 反例 $x_n = n$ ($\arctan x_n \rightarrow \frac{\pi}{2}$ 收敛而 x_n 发散); ② 反例同上 (单调但不收敛).

③ $\arcsin$ 连续, 收敛数列的连续像收敛; ④ 有界单调必收敛, 再取连续像仍收敛.

故 ③④ 正确, 选 B.

76. (I) $|A| = (a-1)^2(a+1) = 0 \implies a = 1$ 或 -1 . 条件 " $A^T X = 0$ 的解都是 $\beta^T X = 0$ 的解" 等价于 β 可由 A 的列向量线性表示.

$a = 1$ 时 A 的列全为 $(1, 0, 1)^T$, 而 β 的第二个分量为 $1 \neq 0$, 不可能;

$a = -1$ 时解 $x_1(-1, 0, 1)^T + x_2(1, -2, 1)^T = \beta$ 得 $x_2 = -\frac{1}{2}, x_1 = \frac{3}{2}, b = -2$.

故 $a = -1, b = -2$.

(II) 此时 $|\lambda E - A| = \lambda(\lambda+2)^2, \lambda = 0$ 的特征向量 $(1, 0, 1)^T$; $\lambda = -2$ 时 $A + 2E$ 秩为 1, 特征向量 $(1, -1, 0)^T, (1, 0, -1)^T$. 取

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \quad P^{-1}AP = \text{diag}(0, -2, -2).$$

77. 由对称性, $\iint_{[0,1]^2} f(x)f(y)dxdy = A^2$ 被对角线分成两个相等的部分, 故

$$I = \frac{A^2}{2}.$$

78. 设 $u = x - y$, 则约束为 $z^2 = u^2 - 1$ ($|u| \geq 1$), 而 $x^2 + y^2 \geq \frac{(x-y)^2}{2} = \frac{u^2}{2}$ (等号在 $x = -y$ 时成立),

$$d^2 = x^2 + y^2 + z^2 \geq \frac{u^2}{2} + u^2 - 1 = \frac{3}{2}u^2 - 1 \geq \frac{1}{2},$$

在 $u = \pm 1, z = 0, x = -y = \pm \frac{1}{2}$ 处取等. 故最短距离为 $\frac{\sqrt{2}}{2}$.

79. 取对数: $\lim_{x \rightarrow 0} \frac{1}{x} \cdot \frac{1 - \cos f(x)}{\sin x} = 1$. 由 $f(0) = 0$ 及 $f'(0)$ 存在,

$$1 - \cos f(x) \sim \frac{f^2(x)}{2}, \quad \frac{f^2(x)}{2x^2} \rightarrow \frac{[f'(0)]^2}{2} = 1,$$

故 $f'(0) = \pm\sqrt{2}$.

80. (I) $A = 2J - E$ (J 为全 1 矩阵), 特征值 $\lambda_1 = 5$ (对应 $(1, 1, 1)^T$), $\lambda_2 = \lambda_3 = -1$ (对应 $x_1 + x_2 + x_3 = 0$, 如 $(1, -1, 0)^T, (1, 0, -1)^T$).

全部特征向量: $k(1, 1, 1)^T$ ($k \neq 0$) 与 $k_1(1, -1, 0)^T + k_2(1, 0, -1)^T$ (不全为零).

(II) $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$, $P^{-1}AP = \text{diag}(5, -1, -1)$.

(III) 施密特正交化并单位化:

$$Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}, \quad Q^{-1}AQ = \text{diag}(5, -1, -1).$$

81. 解空间维数 $= 4 - 2 = 2$; 而 $\alpha_3 = 2\alpha_1 - 3\alpha_2$, 故 α_1, α_2 是一组基. 施密特正交化:

$$\beta_1 = (1, 1, 2, 3)^T, \quad \beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)}\beta_1 = \frac{2}{3}(-2, 1, 5, -3)^T,$$

单位化得标准正交基

$$e_1 = \frac{1}{\sqrt{15}}(1, 1, 2, 3)^T, \quad e_2 = \frac{1}{\sqrt{39}}(-2, 1, 5, -3)^T.$$

82. 下侧时 $d\mathbf{S} = (z'_x, z'_y, -1)dxdy = (2x, 2y, -1)dxdy$, 故

$$I = \iint_{x^2+y^2 \leq 1} [2x^3 - (x^2 + y^2)]dxdy = 0 - \frac{\pi}{2} = -\frac{\pi}{2}.$$

83. $P\{X > 3\} = e^{-3\lambda}$, $1 - (1 - e^{-3\lambda})^3 = \frac{26}{27} \implies (1 - e^{-3\lambda})^3 = \frac{1}{27}$,

即 $1 - e^{-3\lambda} = \frac{1}{3}$, $e^{-3\lambda} = \frac{2}{3}$, 故 $\lambda = \frac{1}{3} \ln \frac{3}{2}$.